Will AI Take Our Jobs?

A Data-Driven Investigation into How Generative AI is Reshaping Work, Skills, and Information Consumption

An Exploratory Analysis Across 5 Data Sources (2020–2026)

Google Trends · FRED · Stack Overflow · LinkedIn · BLS Salary Data

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Note: This report analyses measurable signals from public datasets. Data limitations are acknowledged throughout — the goal is directional insight, not precise prediction. November 2022 (ChatGPT launch) is the primary before/after reference point across all analyses.

1. Executive Summary

This report investigates what measurably changed after generative AI went public in **November 2022** — in how people find information, what skills employers demand, how developers work, and what the labour market is signalling. Five independent data sources are combined to build a before/after picture of the AI shift.

Key Findings

- **AI search interest exploded from near-zero to dominant** in under 24 months. ChatGPT search interest went from 0 to a normalised score of 100 by mid-2025, while traditional platform searches remained flat by comparison.
- **Stack Overflow was overtaken by ChatGPT** as a search destination almost immediately after the November 2022 launch — a direct signal that developers shifted from reading documentation to asking AI.
- **Developer AI adoption jumped from 43.8% to 78.5%** between 2023 and 2025, and resistance ("no and don't plan to") effectively disappeared.
- **AI Engineer job searches surged past Data Analyst** as the most-searched role by 2026 — the vocabulary of the job market changed.
- **ML Engineer and AI Engineer roles demand 20–40x more AI keywords** in job descriptions than traditional analyst roles (LinkedIn, April 2024).
- **Traditional skills did not die — they grew.** Every language in the top 15 increased in developer usage from 2023 to 2025. AI augments, it does not replace.
- **Data Engineer salaries grew 37% from 2020 to 2022** — the strongest pre-ChatGPT salary signal in the dataset.

2. Methodology & Data Limitations

Data Sources

Five independent public datasets are used, each answering a different layer of the central question:

Source	Coverage	What it tells us
Google Trends	Jan 2020 – Jun 2026	Public interest in AI terms vs traditional tools
FRED	Jan 2018 – May 2026	Tech sector employment, GDP, productivity
Stack Overflow	2023 & 2025 only	Developer tool adoption, AI usage, compensation
LinkedIn Postings	April 2024 snapshot	What employers demand in job descriptions
BLS Salary Data	2020 – 2022	Pre-ChatGPT salary baseline by role

The Before/After Framework

All analyses use **November 2022** (ChatGPT public release) as the primary reference point. Data before this date is labelled `pre_chatgpt` and after is labelled `post_chatgpt`. A master timeline parquet file joins all monthly sources into a single 78-row table (one row per month, Jan 2020 – Jun 2026).

Known Limitations

Limitations to keep in mind throughout this report:

- **LinkedIn is April 2024 only** — it is a cross-sectional snapshot, not a trend. We cannot show how employer demand changed over time from this source.
- **Stack Overflow covers only 2023 and 2025** — there is no 2020, 2021, 2022, or 2024 survey data. Before/after comparison uses these two years as proxies.
- **BLS salary data covers only 2020–2022** — it establishes a pre-ChatGPT baseline but cannot show post-AI salary effects.
- **Google Trends values are batch-normalised 0–100** — values within the same batch are comparable, but values across different keyword batches are not.
- **AI Engineer/ML Engineer salary figures are unreliable** — fewer than 75 observations in the BLS dataset. Treat directional only.
- **This project is for learning, not prediction.** The goal is to practise end-to-end data analysis across multiple sources. Exact values matter less than the analytical process and directional findings.

3. Analysis 1 — The Moment Everything Changed

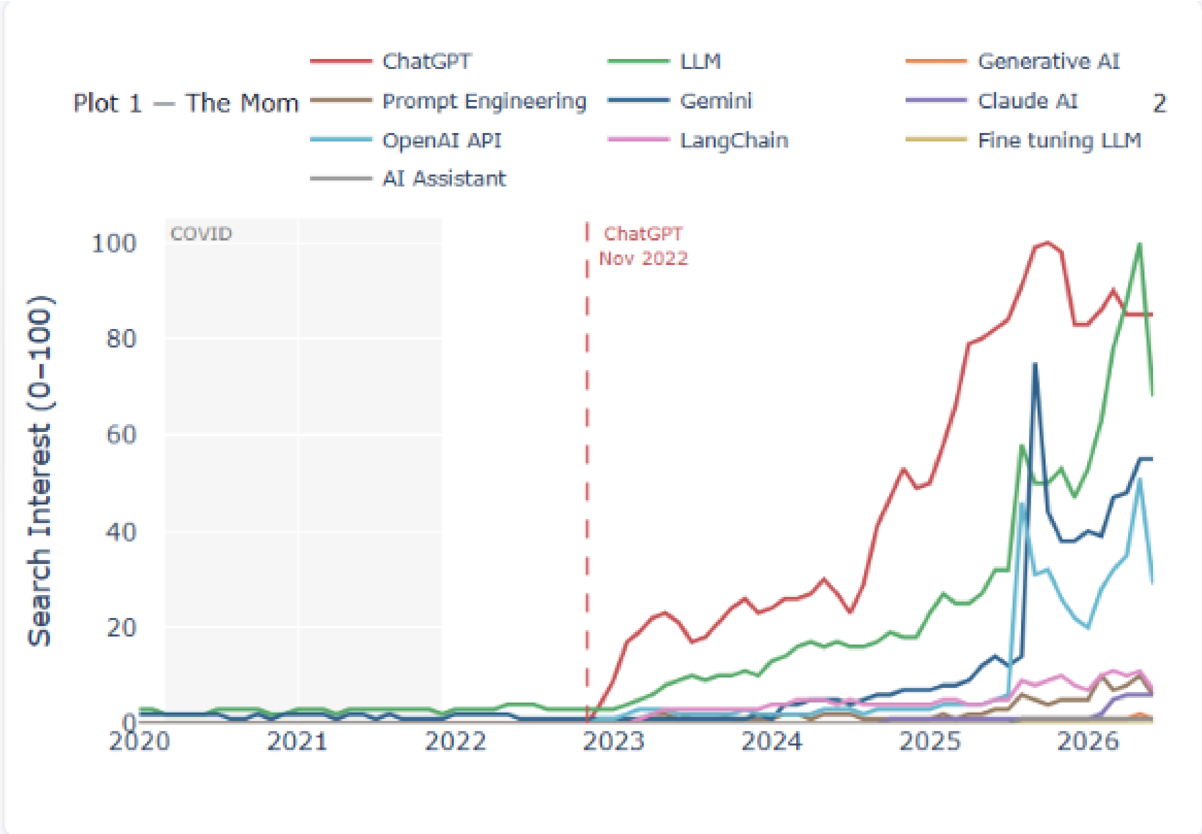

Figure 1: *Google Trends search interest (0–100) for key AI terms, Jan 2020 – Jun 2026. Dashed red line marks ChatGPT public release, November 2022.*

Before November 2022, every AI-related keyword in this chart sits at or near zero. ChatGPT, LLM, Generative AI, Prompt Engineering — none of them registered meaningful public interest. After the ChatGPT launch, interest climbed steadily: ChatGPT (red) reached a normalised score of **100 by mid-2025**, making it the dominant search term in this group. LLM (green) followed a similar trajectory. The sharp Gemini spike around early 2025 corresponds to the Gemini 2.0 launch moment.

Notably, Claude AI, Generative AI, and Prompt Engineering all remain relatively flat even post-ChatGPT. **The public latched onto brand names, not technical concepts.**

> **Key Insight** The AI shift was not gradual. It was a step change. Two years of near-zero interest, then a vertical climb that has not reversed. This is the foundational signal that everything else in this report builds on.

4. Analysis 2 — Are People Still Googling?

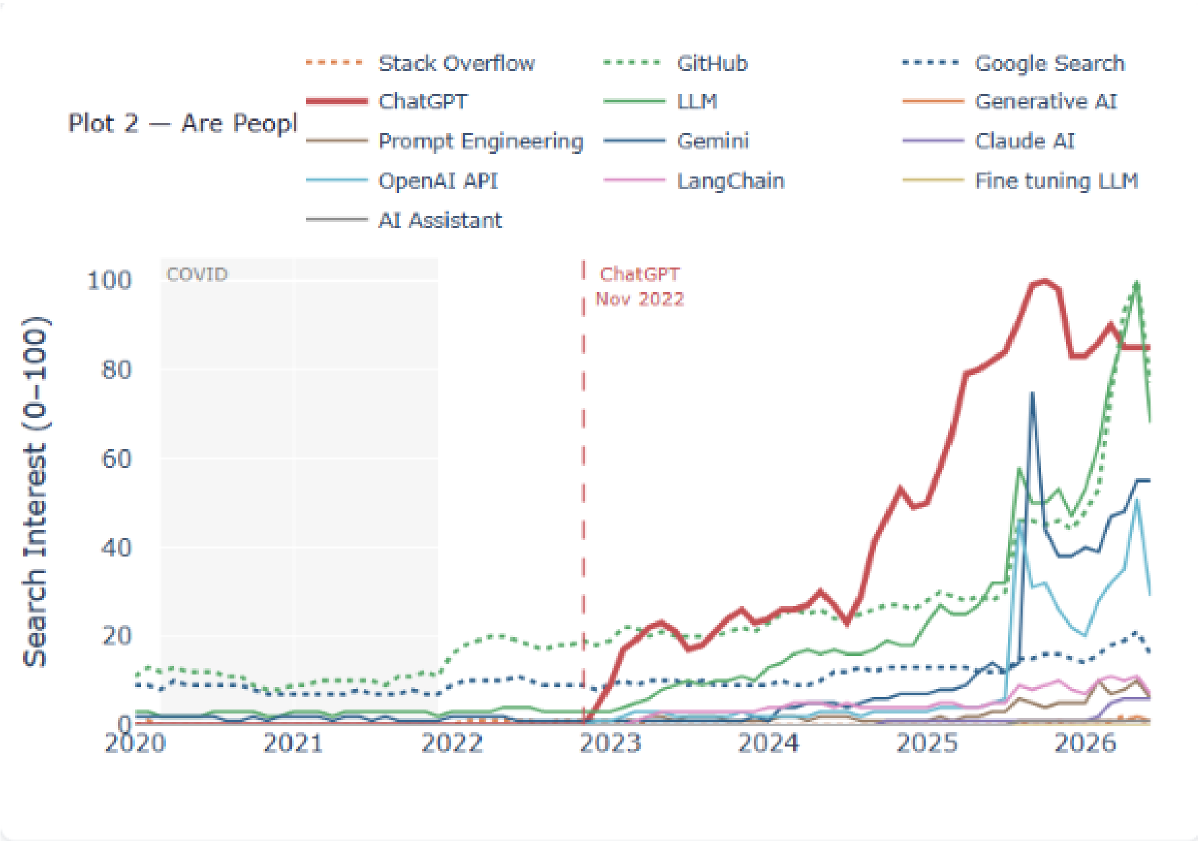

Figure 2: *Traditional platform searches (dotted) vs AI keyword searches (solid), Google Trends 2020–2026. COVID period shaded.*

The three platform lines — Stack Overflow (dotted orange), GitHub (dotted green), and Google Search (dotted blue) — remain completely flat across the entire six-year period. Meanwhile, ChatGPT and LLM surge past them.

The finding here is not that people stopped using Google. It is that **AI search interest grew so rapidly it dwarfed everything else in comparison**. Traditional platforms did not decline in absolute terms — they simply did not grow, while AI tools captured all the new attention.

> **Key Insight** In a world of finite attention, flat is the new declining. While Google, GitHub, and Stack Overflow held steady, all the growth in information-seeking behaviour since 2023 has gone to AI tools. The share of attention shifted even if absolute usage did not.

5. Analysis 3 — Did People Stop Reading Docs?

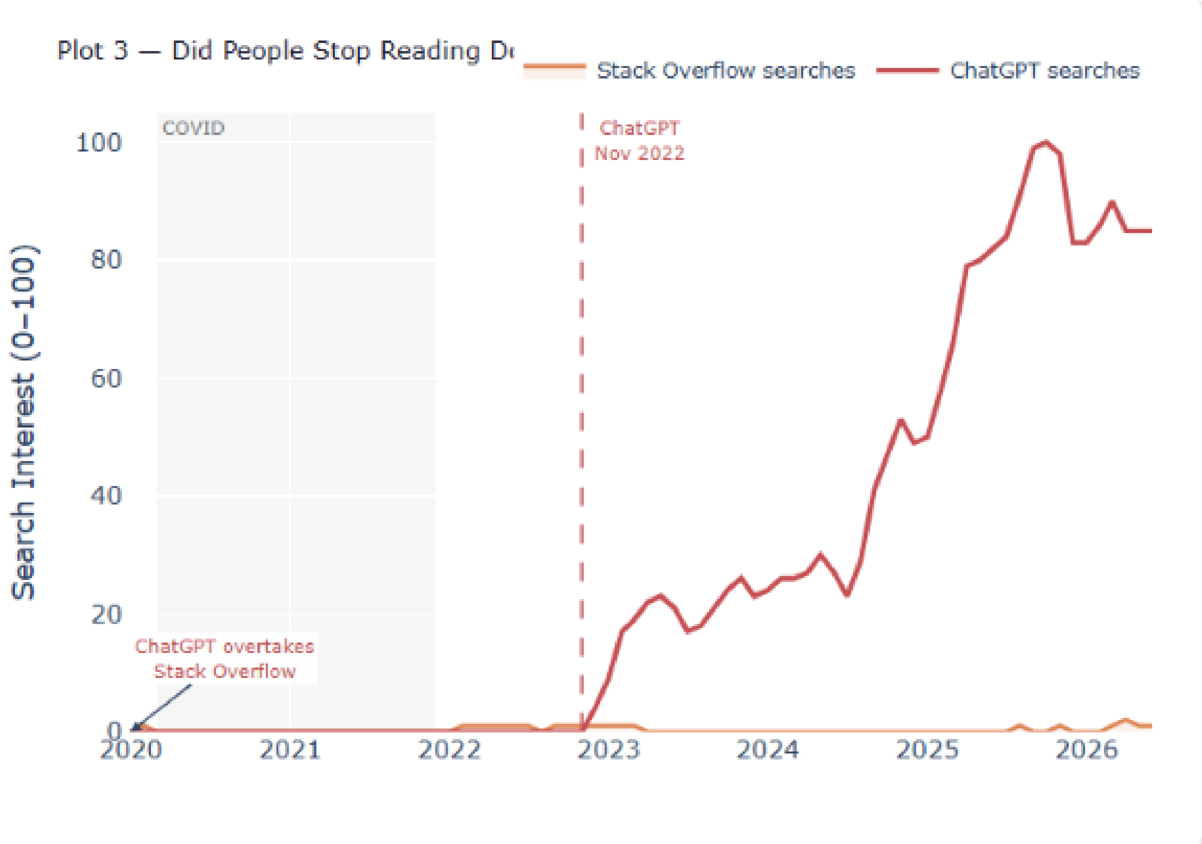

Figure 3: *Stack Overflow search interest (area, orange) vs ChatGPT search interest (line, red). ChatGPT overtakes annotation marks the crossover point.*

Stack Overflow search interest (orange area) stays near zero throughout the entire period, while ChatGPT (red line) climbs from zero to near 100. The "ChatGPT overtakes Stack Overflow" annotation fires almost immediately at the November 2022 launch line.

Important caveat: Stack Overflow's near-zero value is partly a **batch normalisation artefact**. Google Trends normalises each keyword pull independently to 0–100, so Stack Overflow and ChatGPT are not on the same absolute scale. The directional story is valid — ChatGPT interest exploded while Stack Overflow interest stagnated — but the crossover annotation should be interpreted as indicative rather than precise. Stack Overflow also laid off 28% of its staff in 2023, consistent with this directional signal.

Key Insight Developers have shifted from searching for answers on documentation sites to asking AI directly. This is one of the most significant behavioural changes captured in this dataset — and it happened almost overnight.

6. Analysis 4 — Job Search Intent Shifted

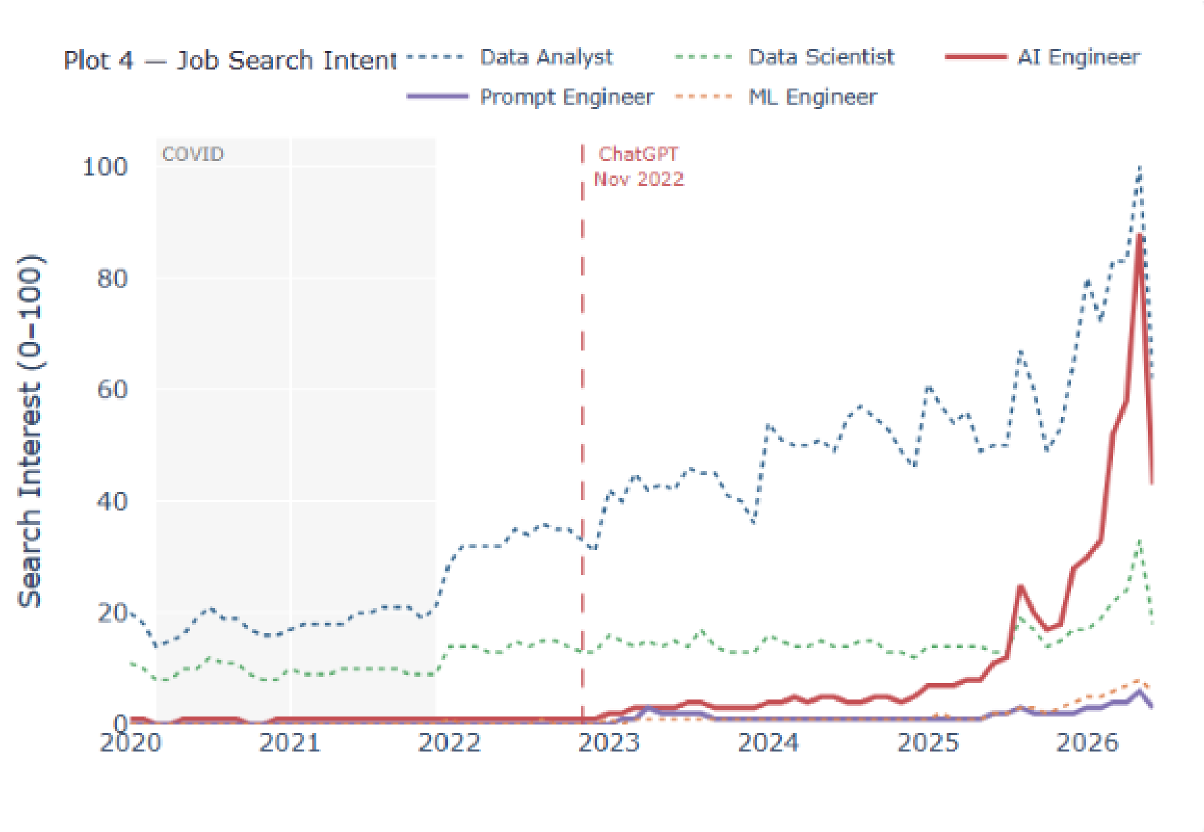

Figure 4: *Google Trends job search interest by role, 2020–2026. AI Engineer and Prompt Engineer shown in solid lines; traditional roles in dotted.*

Before 2022, Data Analyst (blue dotted) was the dominant job search, sitting consistently around 15–20. After November 2022, AI Engineer (solid red) begins climbing and by 2025–2026 overtakes Data Analyst as the most-searched tech role. Data Scientist (green) also grows steadily. Prompt Engineer (purple) barely registers — never breaking above 5 across the entire period.

The 2026 spike in AI Engineer searches is very sharp and may partially reflect normalisation effects of recent data, but the directional trend is clear across multiple years.

Key Insight The vocabulary of job searching changed. People started looking for roles that did not meaningfully exist before 2022. AI Engineer went from near-zero to the most-searched tech role in under three years — one of the fastest job category emergence events in recent search history.

7. Analysis 5 — What Employers Now Want

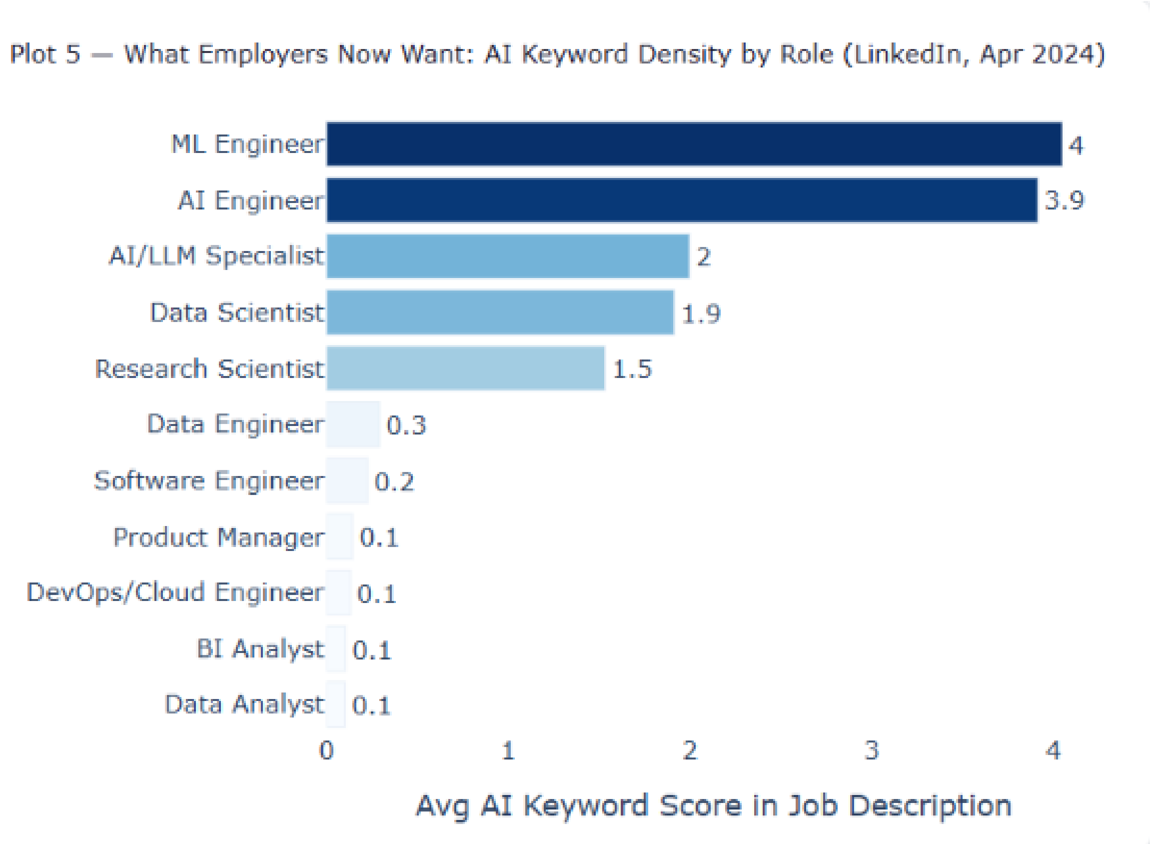

Figure 5: *Average AI keyword score in LinkedIn job descriptions by role category, April 2024 snapshot. Higher score = more AI terms in the job description.*

The divide between roles is stark. ML Engineer (4.0) and AI Engineer (3.9) have dramatically higher AI keyword density than everything else. AI/LLM Specialist (2.0) and Data Scientist (1.9) sit in a mid-tier. Then a sharp cliff — Data Engineer (0.3), Software Engineer (0.2), and everyone else at 0.1.

This reveals **two distinct classes of tech roles** as of April 2024: those where AI is the entire job description, and those where AI barely appears at all. Data Analyst at 0.1 is particularly notable — the most popular data role in search terms (Analysis 4) has almost no AI requirement in actual job postings yet.

> **Key Insight** There is a growing gap between "AI-native" roles (where AI is the job) and "AI-adjacent" roles (where AI has not yet penetrated the job description). As of April 2024, traditional analyst roles have not been redefined by AI in the job market — but the trajectory suggests this is a matter of when, not if.

8. Analysis 6 — Developer AI Adoption

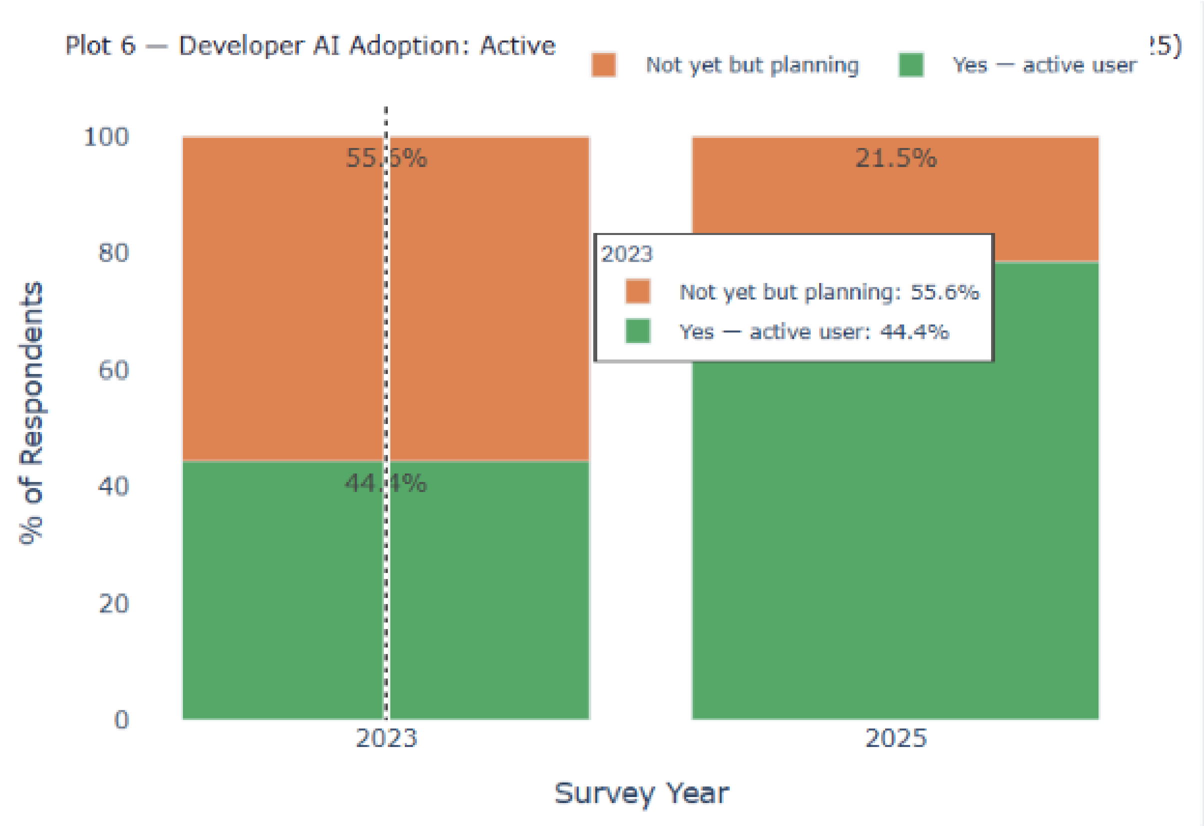

Figure 6: *Developer AI tool adoption, Stack Overflow Survey 2023 vs 2025. Stacked bars show percentage of respondents in each adoption category.*

This is the most dramatic number in the entire project. In 2023, 44.4% of developers were active AI tool users. By 2025, that figure had risen to **78.5%**. The "Not yet but planning" category shrank from 55.6% to 21.5%. Most strikingly, the "No and don't plan to" category has **effectively disappeared** — it is not even visible in the 2025 bar.

> **Key Insight** In just two survey cycles (2023 to 2025), active AI adoption among developers went from a minority to an overwhelming majority. Resistance did not just decline — it nearly vanished. This is adoption at a speed that has no modern precedent in developer tooling. The question is no longer whether developers use AI; it is which tools, for what tasks, and how deeply.

9. Analysis 7 — The Salary Premium

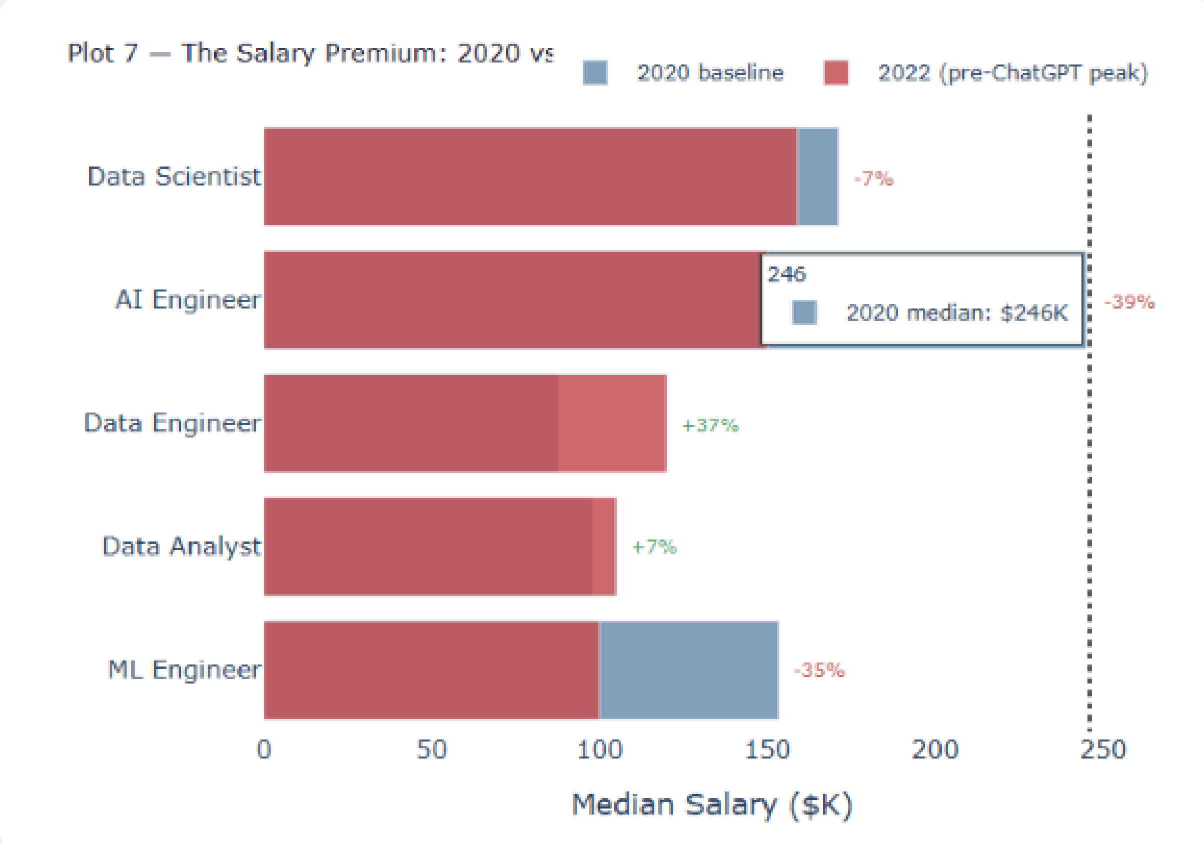

Figure 7: *Median salary by role, 2020 baseline (blue) vs 2022 pre-ChatGPT peak (red). Percentage change annotated. BLS data — see limitation note below.*

Data Engineer shows the strongest growth at **+37%** from 2020 to 2022. Data Analyst grew **+7%**. Data Scientist and AI Engineer show apparent declines (-7% and -39% respectively), which is counterintuitive but explained by data quality.

Reliability warning: The AI Engineer and ML Engineer salary figures in this dataset are based on only 34 and 62 observations respectively — far too few for reliable estimates. The apparent salary declines for these roles are almost certainly a **sample composition artefact** (different job titles being measured across years) rather than a real market signal. The Data Engineer (+37%) and Data Analyst (+7%) figures, based on much larger samples, are more trustworthy. This BLS data only covers 2020–2022 and predates the ChatGPT salary premium entirely.

Key Insight Even before ChatGPT, data infrastructure roles were commanding significant salary growth. The pre-AI era was already rewarding data engineering skills. Post-2022 salary effects are not captured in this dataset — but external reports suggest AI/LLM specialist salaries have grown dramatically since 2023.

10. Analysis 8 — The New Skill Stack

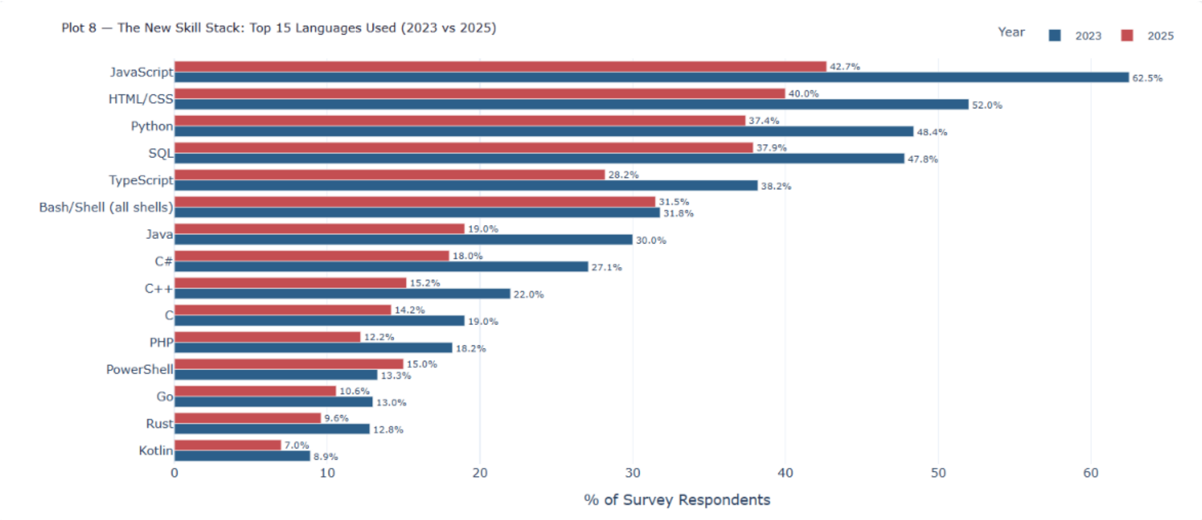

Figure 8: *Top 15 programming languages used by developers, Stack Overflow Survey 2023 (blue) vs 2025 (red). Grouped horizontal bars.*

Every single language in the top 15 grew from 2023 to 2025. Not one declined. JavaScript grew from 42.7% to 62.5% — the largest absolute jump. Python grew from 37.4% to 48.4%. SQL from 37.9% to 47.8%. Even lower-ranked languages like Rust (+3.2pp) and Go (+2.4pp) saw meaningful growth.

The story here is expansion, not substitution. Developers are using more tools, not replacing old ones with new AI-specific ones. The 2025 survey cohort simply uses more languages across the board.

> **Key Insight** AI is not killing traditional programming languages — it is making developers more productive across all of them. The feared "AI will replace coders" narrative is not supported here. Instead, developers who use AI tools appear to be expanding their capability across more languages, not abandoning foundational skills. Python and SQL remain dominant and are *growing* — the most important languages for data work are more in demand in 2025 than in 2023.

11. Conclusion — So, Will AI Take Our Jobs?

The data across all eight analyses tells a nuanced story that is neither the techno-optimist nor the techno-pessimist headline:

1. **AI did not take jobs — it created new ones.** AI Engineer went from near-zero to the most-searched tech role in under three years. The job market expanded its vocabulary, it did not shrink it.

2. **Traditional skills are not dying — they are growing.** Every programming language in the top 15 grew from 2023 to 2025. Python, SQL, and JavaScript are more in demand than ever. AI augments capability; it does not replace foundational skills.

3. **But the gap between AI-native and AI-adjacent roles is widening.** ML Engineers and AI Engineers have 20–40x more AI keywords in their job descriptions than Data Analysts. If this gap does not close, traditional analyst roles risk becoming lower-value in relative terms.

4. **Developer adoption happened at unprecedented speed.** 78.5% of developers actively using AI tools in 2025 (up from 44.4% in 2023) is not a gradual shift. It is a generational adoption event compressed into two years. Resistance has nearly vanished.

5. **Information consumption changed before the job market did.** Developers stopped reading Stack Overflow docs and started asking AI before employers updated job descriptions. The behavioural shift preceded the market signal by at least 12–18 months.

6. **The answer to the title question:** Based on this data — no, AI is not taking jobs. It is *changing* them. The roles that require only pattern-matching and information retrieval are under pressure. The roles that require judgment, multi-source synthesis, and building new things are growing. The people most at risk are not those who do technical work — they are already using AI as a force multiplier. The people most at risk are those who are not adapting.

> **Scope of this analysis:** This report uses 5 public datasets with known limitations (LinkedIn is one month, Stack Overflow is only 2 years, BLS predates ChatGPT). The findings are directional, not definitive. The goal of this project is to demonstrate end-to-end data analysis across multiple heterogeneous sources — building a pipeline, cleaning data, building a dashboard, and communicating findings professionally. The analytical process is the deliverable, not a precise answer to an unanswerable question.

12. References & Data Sources

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